

Lab:15.1

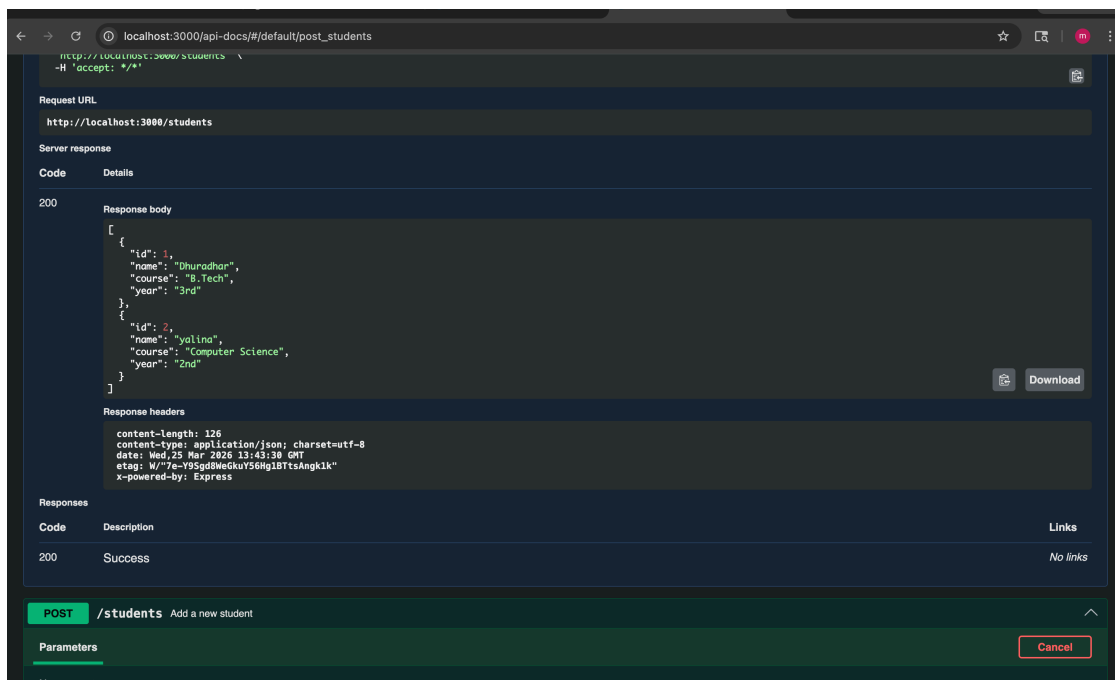
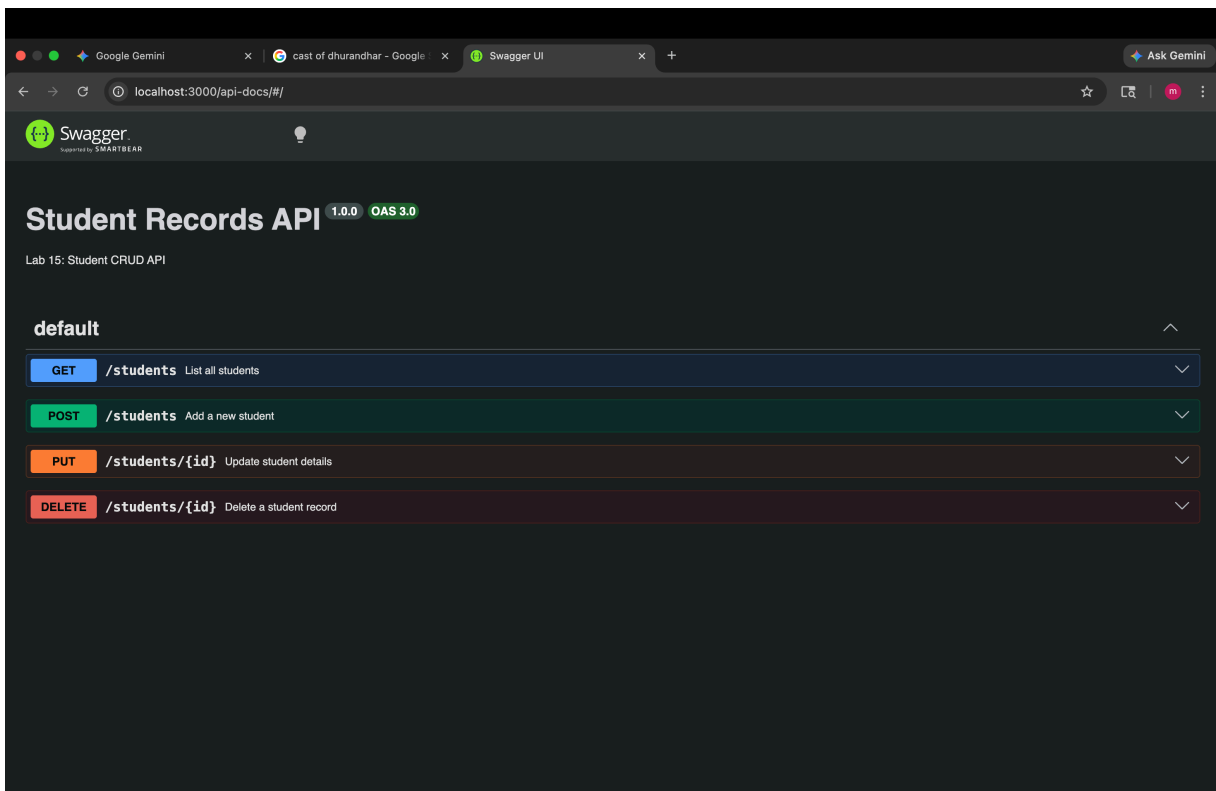
Name: Manupati Eshwar

H.No: 2303A51495

Batch-08

Backend API Development: Creating RESTful services with AI

Task1: Student Records API



Google Gemini x cast of dhurandhar - Google x Swagger UI x + Ask Gemini

localhost:3000/api-docs/#/default/post_students

Parameters Cancel Reset

No parameters

Request body required application/json

Edit Value | Schema

```
{
  "name": "Disha",
  "course": "B-Tech",
  "year": "3rd year"
}
```

Execute

Responses

Code	Description	Links
201	Created	No links

PUT /students/{id} Update student details

Google Gemini x cast of dhurandhar - Google x Swagger UI x + Ask Gemini

localhost:3000/api-docs/#/

Request URL

http://localhost:3000/students

Server response

Code	Details
200	Response body <pre>[{ "id": 1, "name": "Dhurandhar", "course": "B.Tech", "year": "3rd" }, { "id": 2, "name": "yalina", "course": "Computer Science", "year": "2nd" }, { "id": 3, "name": "Disha", "course": "B-Tech", "year": "3rd year" }, { "id": 4, "name": "Disha", "course": "B-Tech", "year": "3rd year" }]</pre> Response headers <pre>connection: keep-alive content-length: 246 content-type: application/json; charset=utf-8 date: Wed, 25 Mar 2026 13:45:27 GMT etag: W/"f6-tV/081jy0V9eE1M/t8hBeFiirk" keep-alive: timeout=5 x-powered-by: Express</pre> Download

Responses

Code	Description	Links
200	Success	No links

Google Gemini x cast of dhurandhar - Google x Swagger UI x + Ask Gemini

localhost:3000/api-docs/#/default/put_students__id_

Responses

Curl

```
curl -X 'PUT' \
  'http://localhost:3000/students/3' \
  -H 'accept: */*'
```

Request URL

http://localhost:3000/students/3

Server response

Code	Details
200	Response body <pre>{ "id": 3, "name": "Disha", "course": "B-Tech", "year": "3rd year" }</pre> Response headers <pre>connection: keep-alive content-length: 59 content-type: application/json; charset=utf-8 date: Wed, 25 Mar 2026 13:46:18 GMT etag: W/"3b-8kr86eI4UcSinxn5NsGAoXf1bGM" keep-alive: timeout=5 x-powered-by: Express</pre>

Responses

Code	Description	Links
200	Updated	No links

DELETE /students/{id} Delete a student record

Google Gemini x cast of dhurandhar - Google x Swagger UI x + Ask Gemini

localhost:3000/api-docs/#/default/delete_students__id_

Responses

Curl

```
curl -X 'GET' \
  'http://localhost:3000/students' \
  -H 'accept: */*'
```

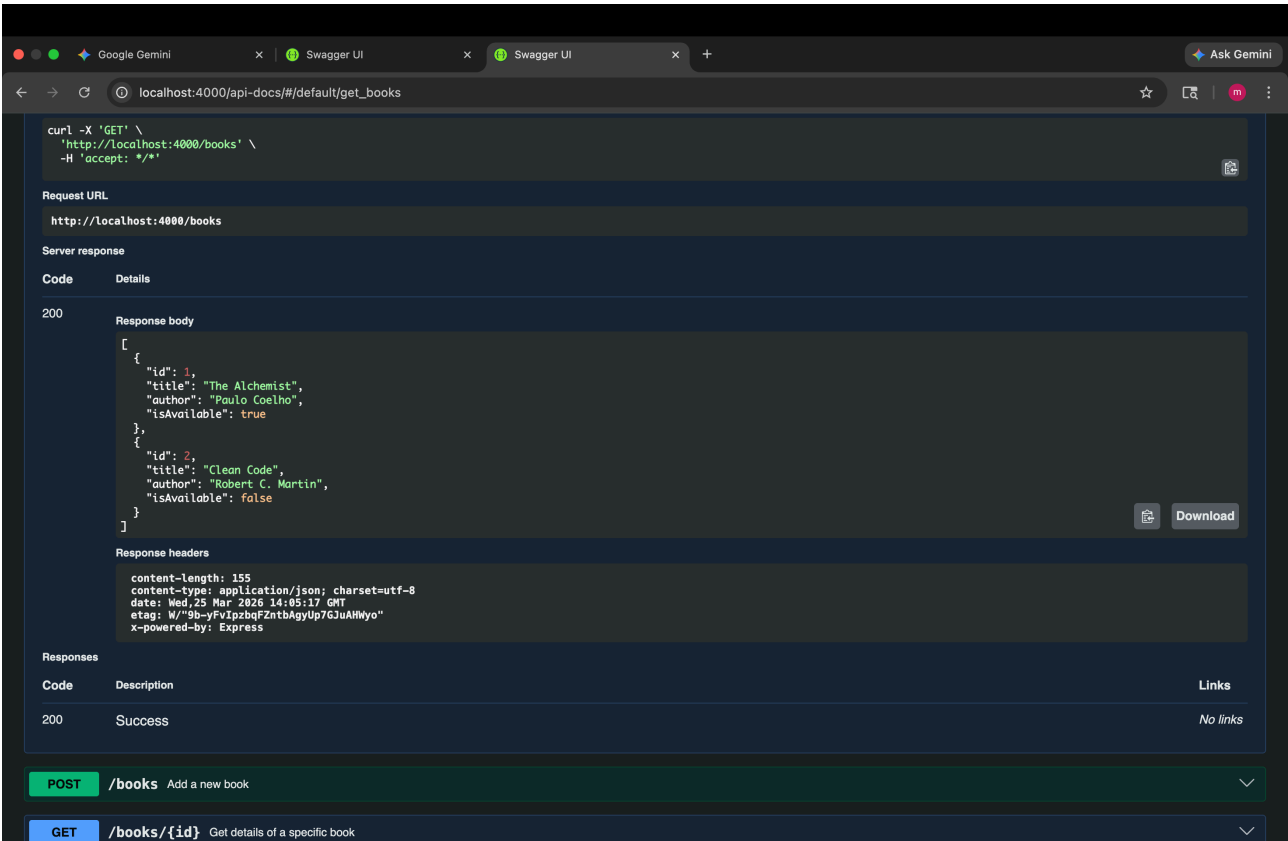
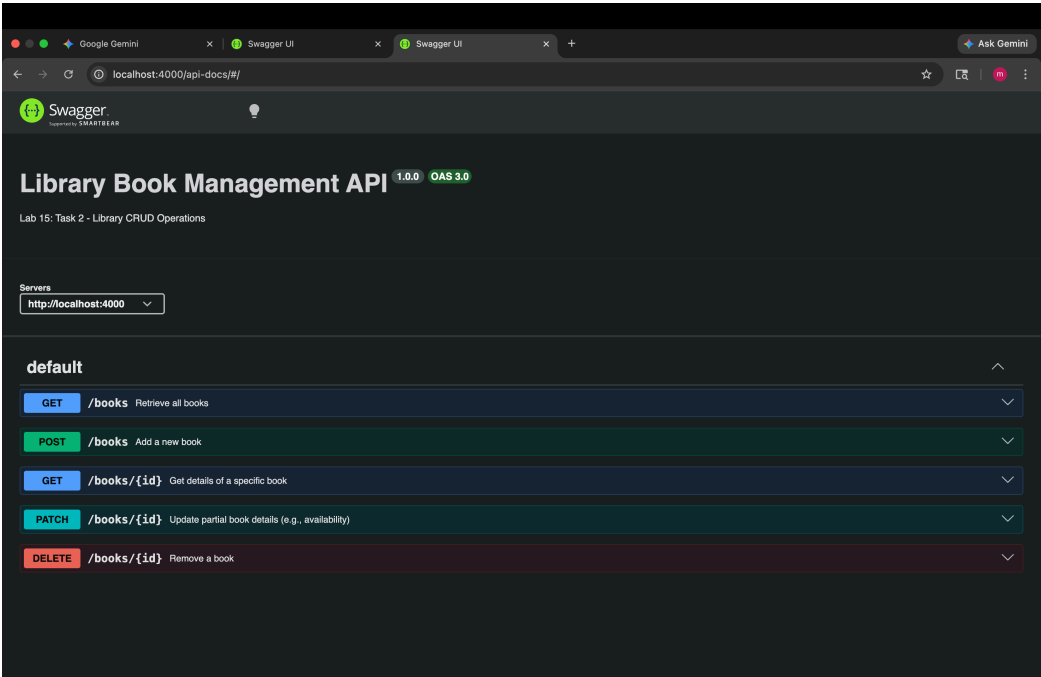
Request URL

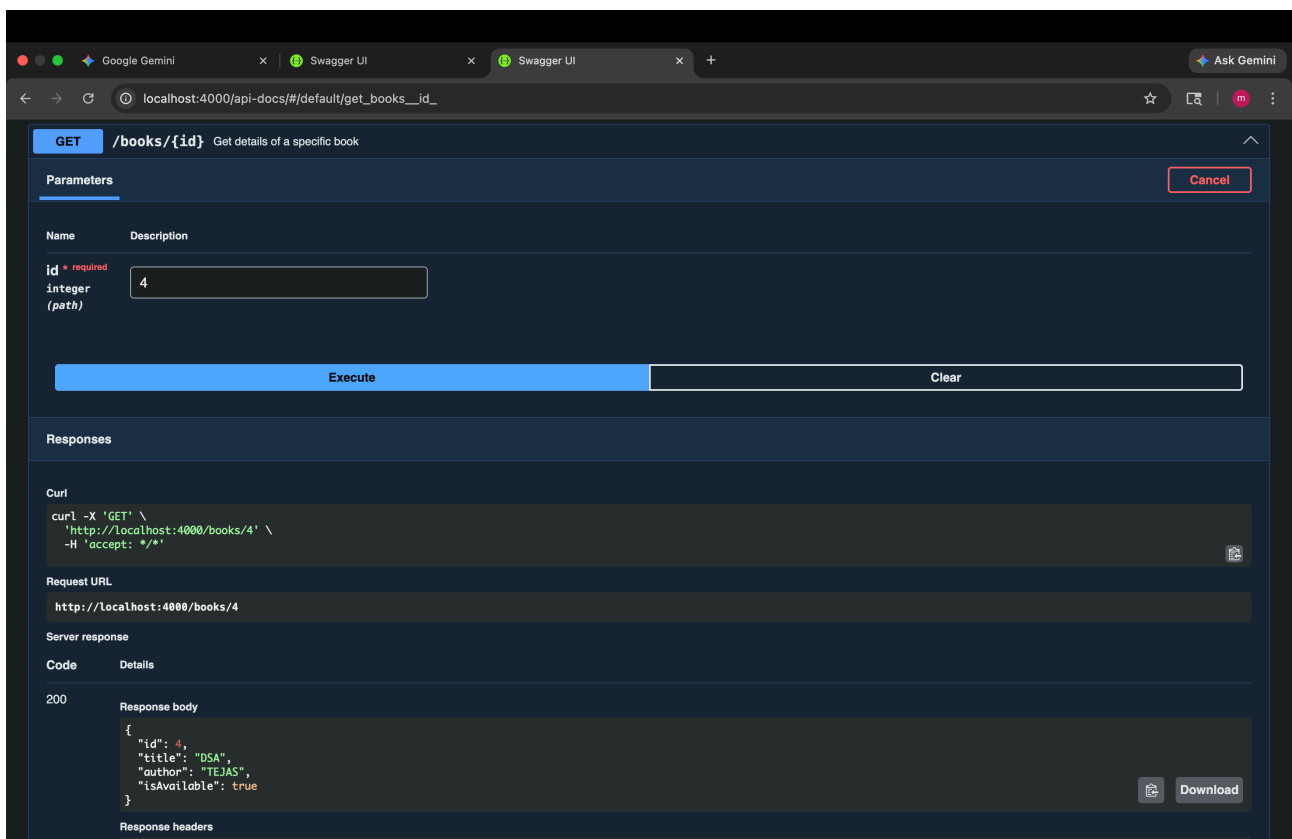
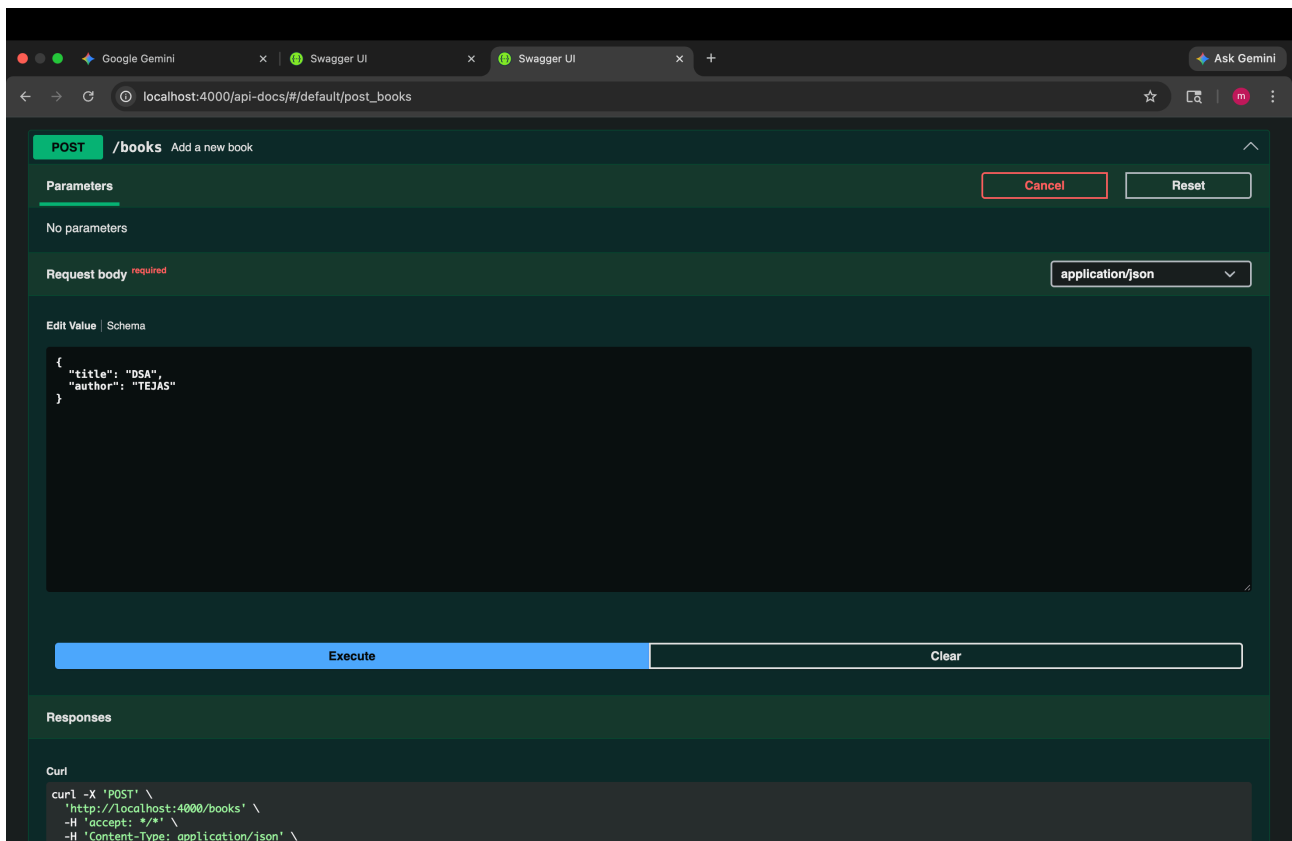
http://localhost:3000/students

Server response

Code	Details
200	Response body <pre>[{ "id": 1, "name": "Dhurandhar", "course": "B.Tech", "year": "3rd" }, { "id": 2, "name": "Yalina", "course": "Computer Science", "year": "2nd" }, { "id": 4, "name": "Disha", "course": "B-Tech", "year": "3rd year" }]</pre> Response headers <pre>connection: keep-alive content-length: 186 content-type: application/json; charset=utf-8 date: Wed, 25 Mar 2026 13:46:57 GMT etag: W/"ba-KHy0HNtmjECVvRIHQknsKBAfzy" keep-alive: timeout=5 x-powered-by: Express</pre>

Task 2:Library Book Management API





Google Gemini x Swagger UI x Swagger UI x + Ask Gemini

localhost:4000/api-docs/#/default/patch_books__id_

Responses

Curl

```
curl -X 'PATCH' \
  'http://localhost:4000/books/1' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
    "isAvailable": true
  }'
```

Request URL

http://localhost:4000/books/1

Server response

Code	Details
200	Response body <pre>{ "id": 1, "title": "The Alchemist", "author": "Paulo Coelho", "isAvailable": true }</pre> Response headers <pre>connection: keep-alive content-length: 75 content-type: application/json; charset=utf-8 date: Wed, 25 Mar 2026 14:08:41 GMT etag: W/"4b-91/0hMGNr0Sa5REtChCbYCLNq3w" keep-alive: timeout=5 x-powered-by: Express</pre>

Responses

Code	Description	Links
200	Updated	No links

DELETE /books/{id} Remove a book

Google Gemini x Swagger UI x Swagger UI x + Ask Gemini

localhost:4000/api-docs/#/default/delete_books__id_

Parameters

Cancel

Name	Description
id * required integer (path)	3

Execute **Clear**

Responses

Curl

```
curl -X 'DELETE' \
  'http://localhost:4000/books/3' \
  -H 'accept: */*' \
```

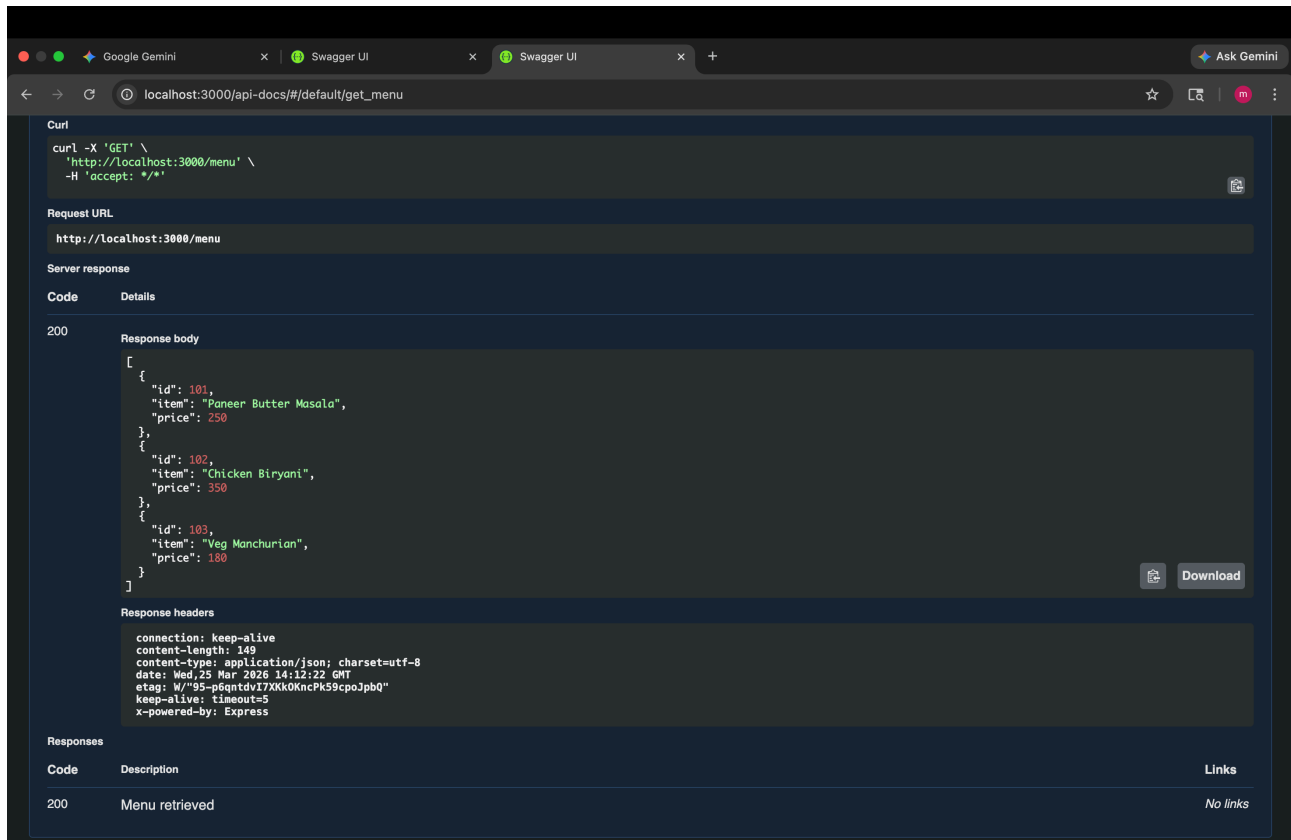
Request URL

http://localhost:4000/books/3

Server response

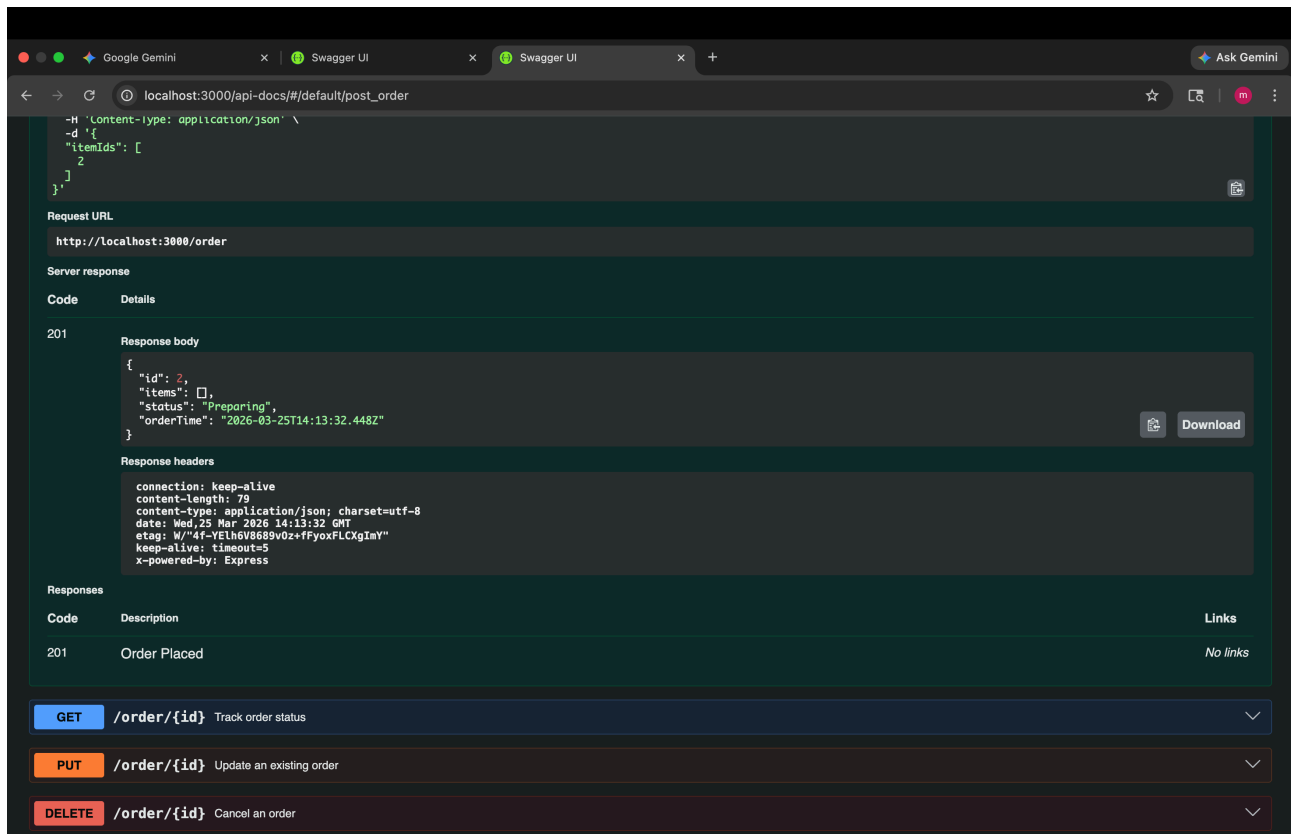
Code	Details
200	Response body <pre>{ "message": "Book deleted" }</pre> Response headers <pre>connection: keep-alive content-length: 26 content-type: application/json; charset=utf-8 date: Wed, 25 Mar 2026 14:09:39 GMT etag: W/"1a-ASXGQesuWor5Uhx4P51EVemPQAU" keep-alive: timeout=5</pre>

Task 4 – Real-Time Application: Online Food Ordering API



This screenshot shows the Swagger UI for the GET /menu endpoint. The browser address bar displays 'localhost:3000/api-docs/#/default/get_menu'. The 'Curl' section contains the command: `curl -X 'GET' \ 'http://localhost:3000/menu' \ -H 'accept: */*'`. The 'Request URL' is 'http://localhost:3000/menu'. The 'Server response' section shows a 200 status code. The 'Response body' is a JSON array of three menu items: `[{"id": 101, "item": "Paneer Butter Masala", "price": 250}, {"id": 102, "item": "Chicken Biryani", "price": 350}, {"id": 103, "item": "Veg Manchurian", "price": 180}]`. The 'Response headers' include: `connection: keep-alive, content-length: 149, content-type: application/json; charset=utf-8, date: Wed, 25 Mar 2026 14:12:22 GMT, etag: W/"95-p6qntdvI7XKk0KncPk59cpoJpbQ", keep-alive: timeout=5, x-powered-by: Express`. The 'Responses' table at the bottom shows a 200 status code with the description 'Menu retrieved' and no links.

Code	Description	Links
200	Menu retrieved	No links



This screenshot shows the Swagger UI for the POST /order endpoint. The browser address bar displays 'localhost:3000/api-docs/#/default/post_order'. The 'Request body' section shows a JSON object: `{ "itemIds": [2] }`. The 'Request URL' is 'http://localhost:3000/order'. The 'Server response' section shows a 201 status code. The 'Response body' is a JSON object: `{ "id": 2, "items": [], "status": "Preparing", "orderTime": "2026-03-25T14:13:32.448Z" }`. The 'Response headers' include: `connection: keep-alive, content-length: 79, content-type: application/json; charset=utf-8, date: Wed, 25 Mar 2026 14:13:32 GMT, etag: W/"4f-YElh6V8689v0z:ffyoXFLCxgImY", keep-alive: timeout=5, x-powered-by: Express`. The 'Responses' table at the bottom shows a 201 status code with the description 'Order Placed' and no links.

Code	Description	Links
201	Order Placed	No links

Below the table, there are three API endpoint definitions:

- GET** /order/{id} Track order status
- PUT** /order/{id} Update an existing order
- DELETE** /order/{id} Cancel an order

Google Gemini x Swagger UI x Swagger UI x + Ask Gemini

localhost:3000/api-docs/#/default/get_order__id_

Parameters

Name	Description
id * required integer (path)	1

Execute Clear

Responses

Curl

```
curl -X 'GET' \
'http://localhost:3000/order/1' \
-H 'accept: */*'
```

Request URL

http://localhost:3000/order/1

Server response

Code	Details
200	Response body <pre>{ "id": 1, "status": "Preparing" }</pre> Response headers <pre>content-length: 29 content-type: application/json; charset=utf-8 date: Wed, 25 Mar 2026 14:14:18 GMT etag: W/"1d-kqH/1u1wxpB6tRUP1r7HJ/1lPI" x-powered-by: Express</pre>

Responses

Google Gemini x Swagger UI x Swagger UI x + Ask Gemini

localhost:3000/api-docs/#/default/put_order__id_

Execute Clear

Responses

Curl

```
curl -X 'PUT' \
'http://localhost:3000/order/1' \
-H 'accept: */*'
```

Request URL

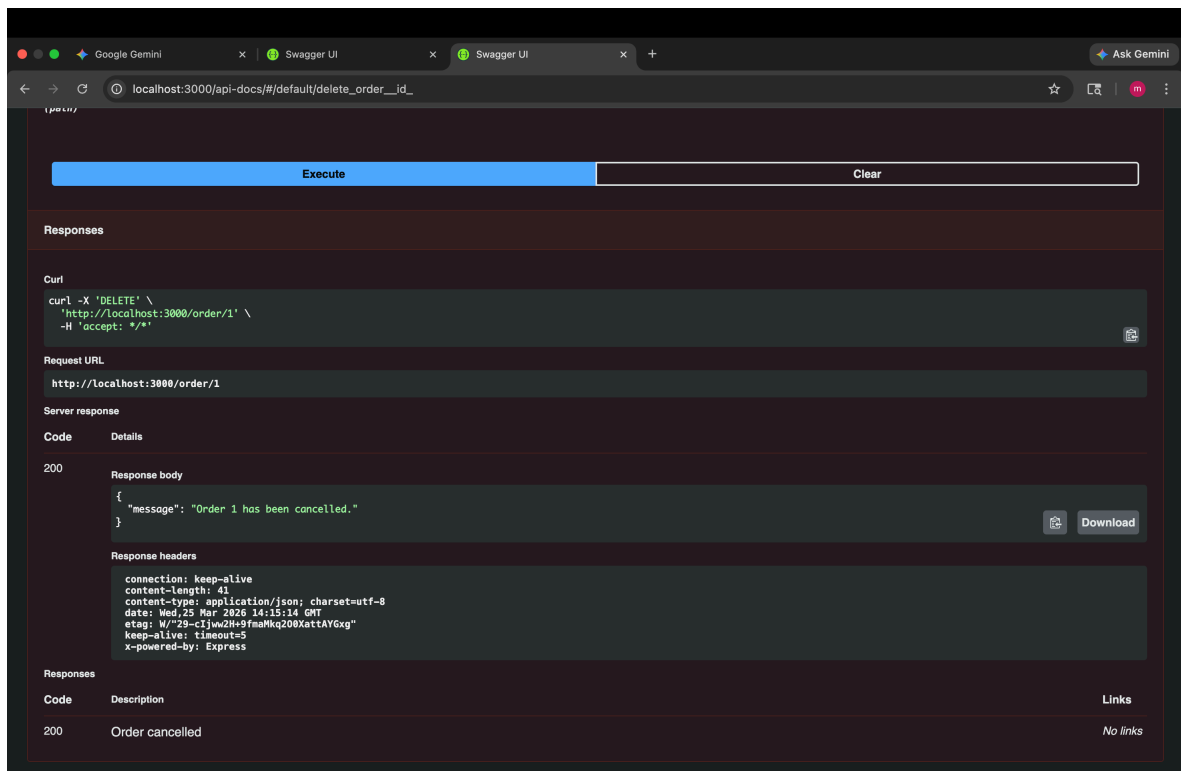
http://localhost:3000/order/1

Server response

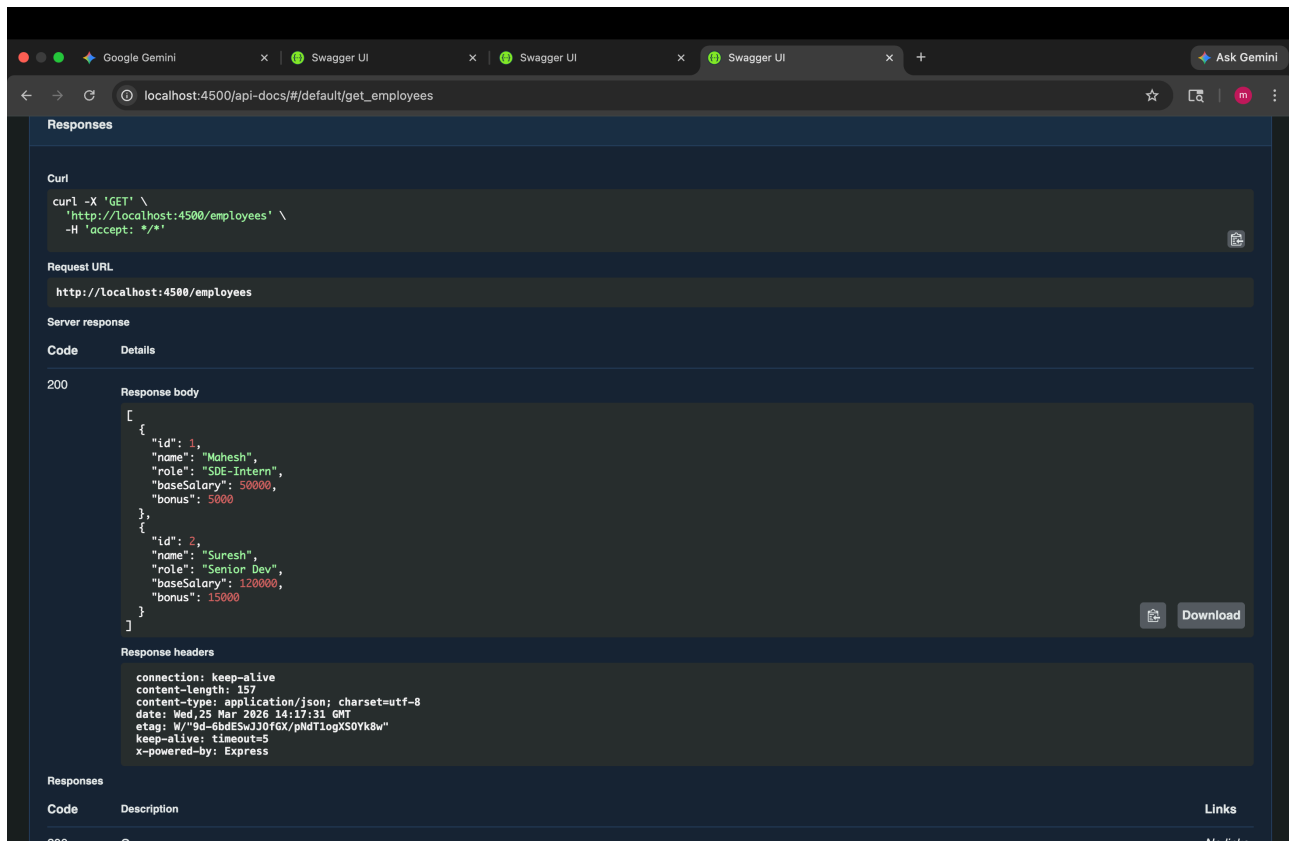
Code	Details
200	Response body <pre>{ "id": 1, "items": [], "status": "Updated/Modified", "orderTime": "2026-03-25T14:13:30.994Z" }</pre> Response headers <pre>connection: keep-alive content-length: 86 content-type: application/json; charset=utf-8 date: Wed, 25 Mar 2026 14:14:43 GMT etag: W/"56-v107w/GW30bVKmbfY31rN0qMR48" keep-alive: timeout=5 x-powered-by: Express</pre>

Responses

Code	Description	Links
200	Order updated	No links



Task3: Employee Payroll API



Google Gemini x Swagger UI x Swagger UI x Swagger UI x Ask Gemini

localhost:4500/api-docs/#/default/post_employees

Responses

Curl

```
curl -X 'POST' \
  'http://localhost:4500/employees' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
    "name": "Naresh",
    "role": "HR",
    "baseSalary": 120000,
    "bonus": 10000
  }'
```

Request URL

http://localhost:4500/employees

Server response

Code	Details
201	Response body <pre>{ "id": 3, "name": "Naresh", "role": "HR", "baseSalary": 120000, "bonus": 10000 }</pre> Response headers <pre>connection: keep-alive content-length: 70 content-type: application/json; charset=utf-8 date: Wed, 25 Mar 2026 14:20:22 GMT etag: W/"46-DN08tS3kh0fcenhA4tzQ8UjKGk" keep-alive: timeout=5 x-powered-by: Express</pre>

Responses

Code	Description	Links
201	Employee Created	No links

Google Gemini x Swagger UI x Swagger UI x Swagger UI x Ask Gemini

localhost:4500/api-docs/#/default/put_employees_id_salary

Responses

Curl

```
curl -X 'PUT' \
  'http://localhost:4500/employees/1/salary' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
    "baseSalary": 5000000,
    "bonus": 100000
  }'
```

Request URL

http://localhost:4500/employees/1/salary

Server response

Code	Details
200	Response body <pre>{ "message": "Salary updated successfully", "employee": { "id": 1, "name": "Mahesh", "role": "SDE-Intern", "baseSalary": 5000000, "bonus": 100000 } }</pre> Response headers <pre>connection: keep-alive content-length: 133 content-type: application/json; charset=utf-8 date: Wed, 25 Mar 2026 14:21:09 GMT etag: W/"85-qhWACs6oha0SS6Q0S+izuYcILfo" keep-alive: timeout=5 x-powered-by: Express</pre>

Responses

Code	Description	Links
------	-------------	-------

Google GeminiSwagger UISwagger UISwagger UIAsk Gemini

localhost:4500/api-docs/#/default/delete_employees_id_

ExecuteClear

Responses

Curl

curl -X 'DELETE' \n'http://localhost:4500/employees/2' \n-H 'accept: */*'

Request URL

http://localhost:4500/employees/2

Server response

Code	Details
200	Response body {\n "message": "Employee 2 removed from system."\n} Download Response headers connection: keep-alive\ncontent-length: 45\ncontent-type: application/json; charset=utf-8\ndate: Wed, 25 Mar 2026 14:21:39 GMT\netag: W/"2d-tdjiBm0HcCUg9/wgy+6hKRPwA8"\nkeep-alive: timeout=5\nx-powered-by: Express

Responses

Code	Description	Links
200	Employee Removed	No links